Implementation Report

Inventory management system- GitHub Repository

Phase 2 focused on implementing the Grocery Shop Inventory Management System using C11 and C++17. The application manages product and supplier records, tracks inventory transactions, generates stock reports, calculates inventory value, and saves inventory data in a text file.

The system is organised into header files (include/), source files (src/), and stored inventory data (data/). C handles the core inventory operations, persistence, and reporting, while C++ provides the application entry point. This modular structure separates the main responsibilities of the application and makes future maintenance easier.

Testing has covered the main functionality, including inventory operations, saving, and reloading data. Further testing should address empty inventories, invalid input, duplicate records, negative quantities, missing files, and malformed data.

Future improvements include:

1. Sample inventory is to be expanded with realistic grocery products such as maize meal, rice, sugar, flour, cooking oil, bread, milk, eggs, and household essentials.
2. The expanded dataset will include appropriate quantities, categories, supplier relationships, reorder thresholds, and clearly labelled illustrative prices in USD.
3. Enhanced searching and sorting, more detailed transaction tracking, low-stock alerts, and additional inventory reports.

Changes are to be implemented incrementally and tested to ensure that existing functionality continues to work correctly.